StreamDCIM: A Tile-based Streaming Digital CIM Accelerator with Mixed-stationary Cross-forwarding Dataflow for Multimodal Transformer

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***Abstract*—Multimodal Transformers are emerging artificial intelligence (AI) models designed to process a mixture of signals from diverse modalities. Digital computing-in-memory (CIM) architectures are considered promising for achieving high efficiency while maintaining high accuracy. However, current digital CIM-based accelerators exhibit inflexibility in microarchitecture, dataflow, and pipeline to effectively accelerate multimodal Transformer. In this paper, we propose StreamDCIM, a tile-based streaming digital CIM accelerator for multimodal Transformers. It overcomes the above challenges with three features: First, we present a tile-based reconfigurable CIM macro microarchitecture with normal and hybrid reconfigurable modes to improve intra-macro CIM utilization. Second, we implement a mixed-stationary cross-forwarding dataflow with tile-based execution decoupling to exploit tile-level computation parallelism. Third, we introduce a ping-pong-like fine-grained compute-rewriting pipeline to overlap high-latency on-chip CIM rewriting. Experimental results show that StreamDCIM outperforms non-streaming and layer-based streaming CIM-based solutions by geomean 2.63× and 1.28× on typical multimodal Transformer models.**

***Index Terms*—digital computing-in-memory (CIM), dataflow, multimodal transformer, reconfigurable architecture.**

I. Introduction

Transformers, a type of neural network (NN) model, have achieved remarkable success across a wide range of artificial intelligence (AI) tasks, outperforming recurrent neural networks (RNNs) and traditional convolution neural networks (CNNs) in both natural language processing (NLP) [1] and computer vision (CV) [2]. Their exceptional performance is largely due to the attention mechanism, which effectively capture contextual knowledge from the entire input sequence. Additionally, a key goal in AI is to emulate human multimodal perception, enabling systems to comprehend information across various modalities, such as language and vision. Recent algorithmic advancements have highlighted the potential of multimodal Transformers in learning from diverse inputs, delivering impressive results in tasks like multilingual image retrieval and action prediction [3], [4].

Abundant prior works have been proposed to accelerate NNs through architectural optimizations. Fig. 1 provides an overview of different NN accelerator architectures. Conventional von Neumann architectures have separate computation and memory units. Some works eliminate the memory access bottleneck by integrating multiply-and-accumulate operations

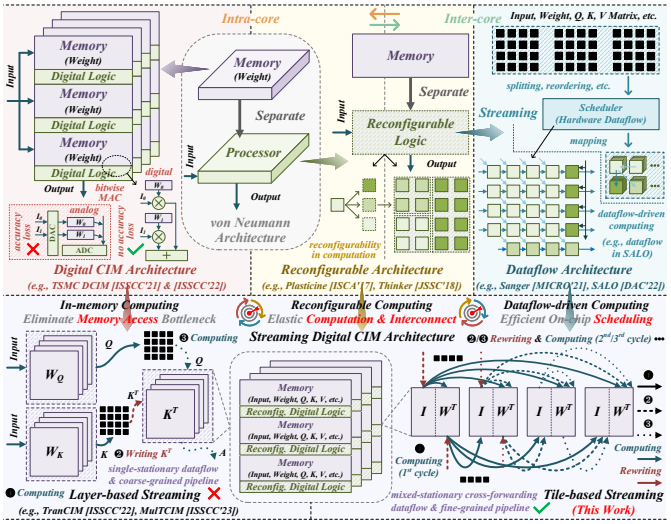


Fig. 1. Overview of Different NN Accelerator Architectures.

directly into memory [5], [6], while others leverage reconfigurable digital logic to enable flexible computation [7], [8]. Additionally, several solutions enhance the efficiency of on-chip scheduling for both computation and memory access through well-designed hardware dataflow [9], [10].

Digital computing-in-memory (CIM) architecture combines the advantages of both CIM and digital architectures, achieving high efficiency by embedding computation directly within memory while ensuring high accuracy by eliminating analog non-ideality through its digital in-memory logic [11]–[14]. Furthermore, unlike analog CIM, digital CIM offers greater customization, allowing for optimization of the in-memory logic. This flexibility makes digital CIM a promising solution to integrate the benefits of in-memory computing, reconfigurable computing, and dataflow-driven computing paradigms, ultimately leading to a streaming digital CIM architecture. Such an architecture holds promise to accommodate efficient memory access, computation and scheduling. The pipeline and parallel reconfigurable modes in TranCIM [12] illustrate layer-based streaming, enhancing overall performance.

However, the multimodal Transformer introduces new challenges for existing digital CIM-based accelerators. Our observations indicate that current digital CIM-based accelerators ex-

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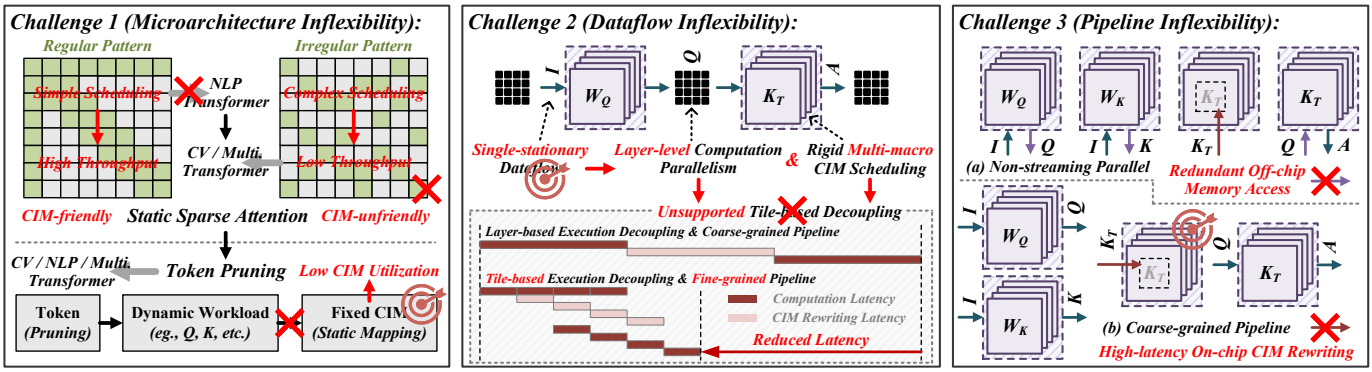


Fig. 2. Challenges for CIM-based Multimodal Transformer Acceleration: 1) Microarchitecture Inflexibility. 2) Dataflow Inflexibility. 3) Pipeline Inflexibility.

hibit ***inflexibility in microarchitecture, dataflow, and pipeline*** to effectively accelerate multimodal Transformer (see Fig. 2).

Challenge 1 (Microarchitecture Inflexibility): Previous digital CIM-based Transformer accelerators mainly rely on static sparse attention, overlooking the benefits of dynamic token pruning, or they lack the flexible microarchitecture for efficient token pruning. Static sparse attention limits attention computations to a predefined set of token pairs, reducing the computational and memory complexity in Transformer models, and is widely used in CIM-based Transformer accelerators [12]–[14]. This approach typically employs a fixed attention pattern for a given task, as it is impossible to predetermine which tokens should be attended to each other [16]–[18]. While static sparse attention is well-suited for NLP applications with long token sequences [19], it often results in irregular patterns in vision and multimodal Transformers due to the complex spatial relationships in images and dynamic token interactions across different modalities. These irregularities hinder achieving high throughput, even with complex scheduling method, particularly for CIM-based Transformer accelerators using rigid weight-stationary dataflows [20], [21]. In contrast, dynamic token pruning offers a more adaptive solution, optimizing tasks in NLP and CV [21], [22]. Multimodal inputs, such as language and images, contain tokens of varying significance [20]–[22]. Token pruning selectively retains attentive tokens and gradually prune inattentive ones across layers. Experimental results show that pruning the redundancy in image tokens can lead to over 1.6× speedup with negligible accuracy loss [21]. However, token pruning introduces dynamic workloads like *Matrix* Q, V, K generation during runtime. In this context, the fixed intra-macro CIM microarchitecture and static workload mapping can result in low CIM utilization.

Challenge 2 (Dataflow Inflexibility): Previous digital CIM-based Transformer accelerators often employ fixed single-stationary dataflows, such as weight-stationary dataflow [12], [13], which are sub-optimal [23], [24] and limited by layer-level computation parallelism and rigid multi-macro CIM scheduling. This limitation restricts support for tile-based execution decoupling and tile-level computation parallelism, ultimately hindering overall performance.

Challenge 3 (Pipeline Inflexibility): Attention layers introduce dynamic matrix multiplications (such as QK^T), where both weights and inputs are generated at runtime. This leads to redundant off-chip memory access for intermediate data in traditional CIM-based Transformer accelerators using non-streaming parallel methods [5], [25], [26]. Although TranCIM [12] mitigates redundant off-chip accesses through pipeline and parallel reconfigurable modes, it still encounters significant pipeline bubbles and incurs additional on-chip CIM rewriting due to its coarse-grained pipeline, resulting in high latency and energy consumption. For example, assuming a 512-bit memory access bandwidth and a benchmark involving QK^T with INT8 precision and a K matrix size of 2048×512, TranCIM incurs over 57% latency to rewrite the K matrix in CIM macros during QK^T computation. When considering Q and K generation, QK^T comprises 66.7% of computations, with CIM rewriting accounting for 88.9% of the latency [15].

In this paper, we propose StreamDCIM, a tile-based streaming digital CIM accelerator for multimodal Transformers. StreamDCIM introduces three levels of flexibility to overcome the above challenges.

1) For **Challenge 1**, we present a tile-based reconfigurable CIM (TBR-CIM) macro ***microarchitecture*** with normal and hrbrid reconfigurable modes to improve the intra-macro CIM utilization.
2) For **Challenge 2**, we implement a mixed-statinary cross-forwarding ***dataflow*** with tile-based execution decoupling and elastic single-macro scheduling to exploit the tile-level computation parallelism.
3) For **Challenge 3**, we introduce a ping-pong-like fine-grained compute-rewriting ***pipeline*** to overlap the high latency of on-chip CIM rewriting.

Experimental results show that StreamDCIM outperforms non-streaming and layer-based streaming CIM-based solutions by geomean 2.63× and 1.28× on typical multimodal Transformer models.

II. StreamDCIM Designs

Multimodal Transformer & Attention Mechanism: The input to an attention layer in vanilla Transformers is typically a sequence of N tokens (I). By multiplying I with weight matrices W_Q, W_K, W_V, we obtain the query (Q), key (K), and value (V) matrices. Next, the attention matrix (A) is obtained by multiplying Q and the transpose of K (K_T). A is

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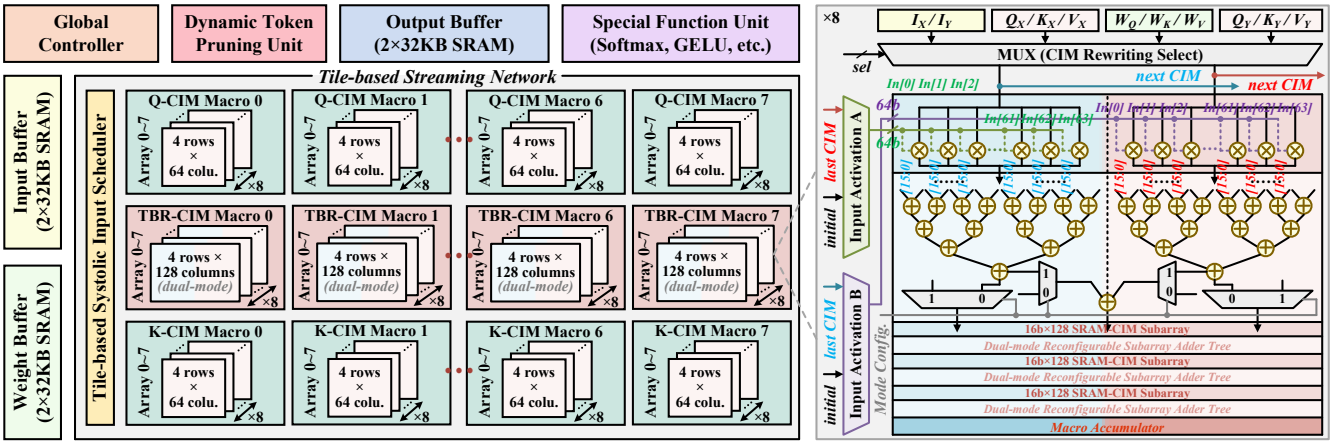


Fig. 3. StreamDCIM: (a) Overall Architecture. (b) TBR-CIM Macro Microarchitecture.

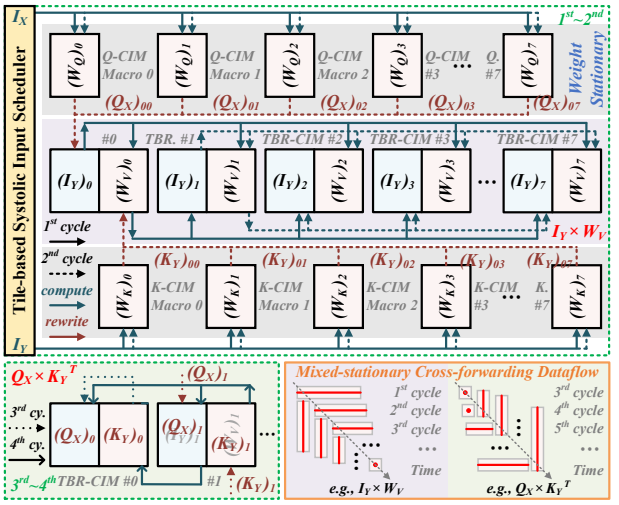
then normalized using the softmax function to yield a probability matrix (P), which is finally multiplied by V to generate the output. In multimodal Transformers, the structure often consists of stacked single-modal and cross-modal encoders to process different modalities, such as vision and language. To process two modalities, the encoders can be divided into two streams for modal X and modal Y. Single-modal attention layers are similar to those in vanilla Transformers. Cross-modal attention layers are introduced to facilitate information exchange between the two modalities. For instance, in the stream for Modal X, Q_X come from modal X ($I_X W_Q$), while K_Y and V_Y are from modal Y ($I_Y W_K$ and $I_Y W_V$). A similar process occurs for modal Y.

The overall architecture of the StreamDCIM accelerator is illustrated in Fig. 3 (a). It comprises a tile-based streaming network (TBSN) with a tile-based systolic input scheduler and three CIM cores: Q-CIM, K-CIM, and TBR-CIM. Additionally, it includes a 64-KB input buffer, a 64-KB weight buffer, a 64-KB output buffer, a dynamic token pruning unit (DTPU), a special function unit (SFU), and a global controller. The CIM cores are interconnected through the TBSN's pipeline bus, with each core containing eight CIM macros.

A. Tile-based Reconfigurable CIM Macro Microarchitecture

Token pruning reduces computational complexity by eliminating redundant token. This process generally includes token ranking and selection, which can leverage attention probabilities, as the probability can indicate a token's relevance to all other tokens [21], [22]. Each token's importance rank can be computed by taking the column mean of its attention probability. Once a token is pruned, its Q, K, V will be excluded from further computations.

To facilitate efficient dynamic token pruning while maintaining high CIM utilization, the DTPU and TBR-CIM macro with normal and hybird reconfigurable modes are proposed, as illustrated in Fig. 3. Similar to Evo-ViT [21] and SpAtten [22], StreamDCIM uses the attention possibility to identify the significance of tokens in various modalities, removing redundant tokens under the control of DTPU. Each TBR-CIM macro consists of eight 4×16b×128 SRAM-CIM arrays and



(a) Mixed-stationary Cross-forwarding Dataflow

	Time	1st cycle	2nd cycle	3rd cycle	4th cycle	5th cycle
TBR-CIM #0		$(I_Y \times W_Y)_0$	$(Q_X \& K_Y^T)_0$	$(Q_X \times K_Y^T)_0$	$(Q_X \times K_Y^T)_1$	$(Q_X \times K_Y^T)_2$
TBR-CIM #1		$(I_Y \times W_Y)_0$	$(I_Y \times W_Y)_1$	$(Q_X \& K_Y^T)_1$	$(Q_X \times K_Y^T)_1$	$(Q_X \times K_Y^T)_2$
TBR-CIM #2		$(I_Y \times W_Y)_0$	$(I_Y \times W_Y)_1$	$(I_Y \times W_Y)_2$	$(Q_X \& K_Y^T)_2$	$(Q_X \times K_Y^T)_2$
TBR-CIM #3		$(I_Y \times W_Y)_0$	$(I_Y \times W_Y)_1$	$(I_Y \times W_Y)_2$	$(I_Y \times W_Y)_3$	$(Q_X \& K_Y^T)_3$
Q-CIM #0-7		$(Q_X)_0$	$(Q_X)_1$	$(Q_X)_2$	$(Q_X)_3$	$(Q_X)_4$
K-CIM #0-7		$(K_Y)_0$	$(K_Y)_1$	$(K_Y)_2$	$(K_Y)_3$	$(K_Y)_4$

(b) Fine-grained Compute-Rewriting Pipeline

Fig. 4. (a) Mixed-stationary Cross-forwarding Dataflow & (b) Fine-grained Compute-Rewriting Pipeline (Example of the Stream for Modal X).

each array has four rows of dual-mode reconfigurable subarray adder trees and one macro accumulator. As token pruning progresses, TBR-CIM macros that initially store both I and W in hybrid mode ($mode_config = 0$) can be reconfigured to operate in normal mode ($mode_config = 1$), storing only W. In normal mode, the TBR-CIM macro functions as a weight-stationary CIM macro, accelerating the Q, K, V generation.

B. Mixed-stationary Cross-forwarding Dataflow

To support the tile-based execution decoupling, the mixed-stationary cross-forwarding dataflow is introduced, as shown in Fig. 4 (a) for an example of modal X. For instance, in the 1st cycle, TBR-CIM #0 provides row-wise I_Y and column-wise W_V as inputs to other CIM macros. Each row from $(I_Y)_0$ is

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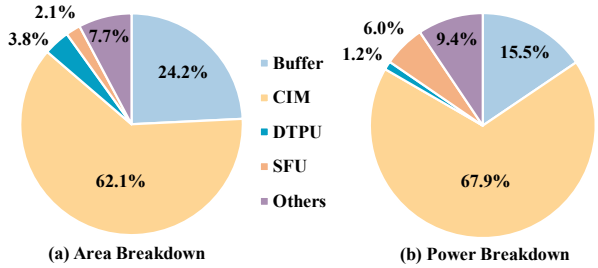


Fig. 5. StreamDCIM: (a) Area Breakdown. (b) Power Breakdown.

sent to the W_V part of TBR-CIM #0-7, generating a full row in V. Concurrently, each column from $(W_V)_0$ is sent to the I_Y part of TBR-CIM #1-7, producing a portion of a column in V. This cross-forwarding computation approach leverages the mixed-stationary TBR-CIM configuration, facilitating more frequent reuse of stored data and utilizing tile-level computation parallelism. Additionally, the calculation of $Q_X K_Y^T$ operates as the inverse of the process for $I_Y W_V$, which also exploits the mixed-stationary cross-forwarding dataflow. Conversely, the calculation of $I_X W_Q$ and $I_Y W_K$ follows the fixed weight-stationary dataflow.

C. Ping-pong-like Fine-grained Compute-Rewriting Pipeline

To overlap the high latency associated with on-chip CIM rewriting, the ping-pong-like fine-grained compute-rewriting pipeline is proposed, as shown in Fig. 4 (b). The above mixed-stationary cross-forwarding dataflow enables tile-based computation parallelism and elastic single-macro CIM scheduling, providing the possibility for a more finer-grained pipeline. In this pipeline, specific resources can be dynamically released and reallocated. For instance, in the 2nd cycle, the inputs and weights stored in TBR-CIM #0 are no longer required for ongoing calculations, freeing TBR-CIM #0 for immediate rewriting. Meanwhile, $(Q_X)_0$ and $(K_Y)_0$ computations have been completed in Q-CIM #0-7 and K-CIM #0-7, respectively. This timing permits the parallel execution of rewriting $(Q_X)_0$ and $(K_Y)_0$ alongside new computations of $(Q_X)_1$, $(K_Y)_1$, and $(V_Y)_1$. Such parallelism effectively overlaps rewriting latency, enhancing throughput by ensuring continuous resource utilization across cycles.

III. Experimental Results

A. Experiment Settings

Methodology. We implement all modules of StreamDCIM using Verilog. We synthesize the design using Synopsys Design Compiler with a commercial 28nm technology, meeting timing at a 200 MHz frequency. We use Synopsys PrimeTime PX for accurate power analysis. To evaluate our design, we select two typical multimodal Transformer models, ViLBERT-base and ViLBERT-large [27], with the visual question answering (VQA) v2.0 dataset. To maintain high accuracy, the attention layers use INT16 precision. In our configuration, *X* is the vision modality, while *Y* is the language modality, each with a token count of $N_X = N_Y = 4096$.

Comparison. We compare StreamDCIM using tile-based streaming solution (**Tile-stream**) with two digital CIM-based accelerators: non-streaming solution and layer-based streaming solution. The non-streaming solution (**Non-stream**) operates similarly to the conventional work mode of previous CIM accelerators [5], [25], [26]. In Non-stream, dynamic matrix multiplication in the attention layers lead to redundant off-chip memory access for intermediate data, which negatively impacts performance and efficiency. The layer-based streaming solution (**Layer-stream**) aligns with the parallel and pipeline reconfigurable mode of TranCIM [12]. In Layer-stream, on-chip CIM rewriting during execution due to layer-based streaming introduces high latency.

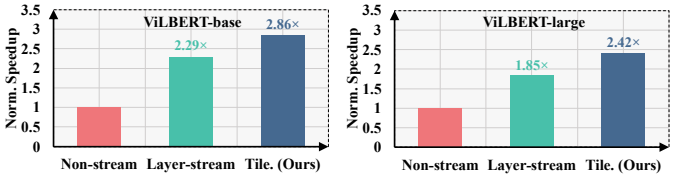


Fig. 6. Performance Comparison on ViLBERT-base and ViLBERT-large.

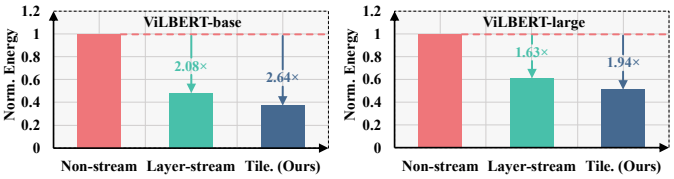


Fig. 7. Energy Comparison on ViLBERT-base and ViLBERT-large.

B. Area and Power Breakdown

Fig. 5 shows the evaluated area and power breakdown of the essential modules in StreamDCIM. Evaluation shows that StreamDCIM has a chip area of 12.10 mm^2 in a 28nm process, and consumes a maximum power of 122.77 mW.

C. Performance and Energy Comparison

Fig. 6 illustrates the performance comparison between StreamDCIM and both non-streaming and layer-based streaming solutions on the ViLBERT-base and ViLBERT-large models. Specifically, StreamDCIM achieves a speedup of 2.86× and 1.25× over Non-stream and Layer-stream, respectively, on ViLBERT-base model. For ViLBERT-large model, StreamDCIM provides a speedup of 2.42× compared to Non-stream and 1.31× over Layer-stream.

Fig. 7 shows the energy comparison, normalized to the non-streaming solution, across the ViLBERT-base and ViLBERT-large models. On ViLBERT-base model, StreamDCIM reduces energy consumption by 2.64× over Non-stream and 1.27× over Layer-stream. On ViLBERT-large model, StreamDCIM achieves energy saving of 1.94× compared to Non-stream and 1.19× over Layer-stream.

IV. Conclusion

In this paper, we describe StreamDCIM, a tile-based streaming digital CIM accelerator with mixed-stationary cross-forwarding dataflow for multimodal Transformers. We implement StreamDCIM using Verilog and demonstrate that comparede with non-streaming and layer-based streaming CIM-based solutions, StreamDCIM attains by geomean 2.63×, 1.28× speedup and 2.26×, 1.23× energy saving on typical multimodal Transformer models.

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